Problem A. Echo Prefixes

Input file: `standard input`
Output file: `standard output`
Time limit: 4 seconds
Memory limit: 512 megabytes

You are given a lowercase English string S of length n. The string S is fixed.

Each position i has a non-negative weight a_i. The weights may be changed.

For a non-empty string X, define its *echo value* as

$$H(X) = \sum_{\substack{1 \le j \le n-|X|+1 \\ S[j..j+|X|-1]=X}} a_j.$$

That is, every occurrence of X in S contributes the current weight of its starting position.

Process q operations:

- `1 i x`: set $a_i = x$.
- `2 p m`: output
$$\sum_{t=1}^{m} H(S[p..p+t-1]).$$
- `3 p m k`: find the minimum t such that $1 \le t \le m$ and
$$\sum_{d=1}^{t} H(S[p..p+d-1]) \ge k.$$
If there is no such t, output 0.

All indices are 1-based.

Input

The first line contains two integers n, q $(1 \le n, q \le 2 \cdot 10^5)$.

The second line contains a lowercase English string S of length n.

The third line contains n integers $a_1, a_2, \ldots, a_n$ $(0 \le a_i \le 10^6)$.

Each of the next q lines contains one operation in one of the following formats:

- `1 i x` $(1 \le i \le n,\ 0 \le x \le 10^6)$
- `2 p m` $(1 \le p \le n,\ 1 \le m \le n-p+1)$
- `3 p m k` $(1 \le p \le n,\ 1 \le m \le n-p+1,\ 1 \le k \le 10^{18})$

It is guaranteed that all output answers do not exceed $9 \cdot 10^{18}$.

Output

For each operation of type 2 or 3, output a single integer on a separate line.

Example

standard input	standard output
5 9 ababa 1 0 2 1 3 2 1 5 3 1 5 10 1 1 5 2 1 5 3 1 5 20 2 2 3 1 4 4 3 2 3 8 2 2 3	14 3 34 3 2 2 8

Note

Initially, $S =$ `ababa` and $a = [1, 0, 2, 1, 3]$.

For $p = 1, m = 5$, the prefixes are `a`, `ab`, `aba`, `abab`, `ababa`, and their echo values are $6, 3, 3, 1, 1$. Thus the first answer is 14.

For the query `3 1 5 10`, the cumulative sums for $t = 1, 2, 3$ are $6, 9, 12$, so the answer is 3.

After setting $a_1 = 5$, the same five echo values become $10, 7, 7, 5, 5$, so the answer to `2 1 5` is 34.

Problem B. Enlarged Badge

Input file: standard input
Output file: standard output
Time limit: 3 seconds
Memory limit: 512 megabytes

You are given two finite sets of lattice points A and B on the plane. Let $P = conv(A)$ and $Q = conv(B)$ be their convex hulls. The input may contain duplicate points.

For a positive integer k, scale P by a factor of k with respect to the origin, obtaining the polygon kP.

An integer vector (u, v) is called valid if the translated polygon $kP + (u, v)$ and Q have at least one common point. Boundary intersection also counts.

For each query k, determine the number of valid integer vectors (u, v) modulo 998244353.

Input

The first line contains three integers n, m, q ($3 \le n, m \le 2 \cdot 10^5$, $1 \le q \le 2 \cdot 10^5$) — the number of points in A, the number of points in B, and the number of queries.

Each of the next n lines contains two integers x_i, y_i ($-10^9 \le x_i, y_i \le 10^9$), describing a point of A.

Each of the next m lines contains two integers x_i, y_i ($-10^9 \le x_i, y_i \le 10^9$), describing a point of B.

Each of the next q lines contains one integer k ($1 \le k \le 10^{18}$), describing a query.

It is guaranteed that both P and Q have positive area.

Output

For each query, output one integer — the number of valid integer vectors modulo 998244353.

Example

standard input	standard output
6 5 4 0 0 2 0 2 1 0 1 1 0 1 1 0 0 3 0 1 2 0 1 1 1 1 2 5 10	20 36 108 308

Note

In the sample, P is the rectangle with vertices $(0, 0), (2, 0), (2, 1), (0, 1)$, and Q is the quadrilateral with vertices $(0, 0), (3, 0), (1, 2), (0, 1)$.

We have $2S_P = 4$, $2S_Q = 7$, $B_P = 6$, $B_Q = 7$, and $2S_{Q+(-P)} = 25$. Hence the answer for query k is $2k^2 + 10k + 8$.

For $k = 1, 2, 5, 10$, the answers are $20, 36, 108, 308$.

Problem C. Number

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 seconds
Memory limit:	256 megabytes

Given a positive integer base B.

You need to construct three permutations P, Q, R of length B (i.e., each sequence contains exactly each number from 0 to $B-1$ once):

$P = (p_1, p_2, \ldots, p_B), \quad Q = (q_1, q_2, \ldots, q_B), \quad R = (r_1, r_2, \ldots, r_B),$

Define the integer represented by a sequence $X = (x_1, x_2, \ldots, x_B)$ in base B as

$$[X]_B = \sum_{i=1}^{B} x_i \cdot B^{B-i}.$$

Note that sequences are allowed to start with digit 0, so the represented integer can be less than B^{B-1}.

Please construct three permutations P, Q, R such that they satisfy the equation (addition is performed in base B)

$$[P]_B + [Q]_B = [R]_B,$$

and for every position i ($1 \le i \le B$), the digits p_i, q_i, r_i are pairwise distinct.

If no such construction exists, output -1.

Input

The input contains a single integer B ($2 \le B \le 10^6$).

Output

If a valid construction exists, output three lines, each containing B integers, representing the sequences P, Q, R respectively, with integers separated by spaces.

If no valid construction exists, output a single line containing -1.

If multiple solutions exist, output any one of them.

Example

standard input	standard output
`5`	`-1`

`6`	`4 2 5 0 3 1`
	`0 4 1 2 5 3`
	`5 1 0 3 2 4`

Note

Problem D. Half-plane Cover

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 seconds
Memory limit:	1024 megabytes

You are given several test cases. In each test case, there are n closed half-planes on the plane.

The i-th half-plane is described by three integers a_i, b_i, c_i and contains all points (x, y) satisfying

$$a_i x + b_i y + c_i \geq 0.$$

For each test case, choose some of the half-planes so that their union is the whole plane. In other words, every point (x, y) must belong to at least one chosen half-plane.

Determine whether this is possible. If it is possible, output a choice with the minimum number of half-planes.

Input

The first line contains an integer t ($1 \leq t \leq 5000$), the number of test cases.

For each test case, the first line contains an integer n ($1 \leq n \leq 5000$), the number of half-planes.

Each of the next n lines contains three integers a_i, b_i, c_i ($-10^6 \leq a_i, b_i, c_i \leq 10^6$), describing one half-plane.

For every i, a_i and b_i are not both equal to 0.

It is guaranteed that the sum of n over all test cases does not exceed 5000.

Output

For each test case, output the answer independently.

If there is no valid choice, output a single line containing `NO`.

Otherwise, output `YES` on the first line.

On the second line, output an integer k, the minimum possible number of selected half-planes.

On the third line, output k pairwise distinct integers, the indices of the selected half-planes in this test case. Indices are 1-based.

If there are several optimal answers, output any of them.

Example

standard input	standard output
3 2 1 0 0 -1 0 0 3 1 0 0 0 1 0 -1 -1 0 3 1 0 -1 0 1 -1 -1 -1 -1	YES 2 1 2 YES 3 1 2 3 NO
1 5 1 0 -100 0 1 -100 1 1 -1000 2 2 -2 -3 -3 6	YES 2 4 5
1 4 -1000000 -1000000 -1000000 1000000 -1000000 1000000 1000000 1000000 -1000000 -1000000 999999 1000000	YES 3 2 3 4

Note

In the first test case, the two half-planes are $x \geq 0$ and $x \leq 0$. Their union is the whole plane.

In the second test case, the three half-planes are $x \geq 0$, $y \geq 0$, and $x + y \leq 0$. No two of them cover the whole plane, but all three do.

In the third test case, the point $(0, 0)$ does not belong to any given half-plane, so it is impossible to cover the whole plane.

The half-planes are closed, so points on the boundary are included.

In sample 2, the last two half-planes cover the whole plane, while the first three are distractors and do not form a smaller valid cover.

Sample 3 stresses exact integer arithmetic near the coefficient limits.

Problem E. Koishi and Function

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 seconds
Memory limit:	256 megabytes

In number theory, a function $f(n)$ is called **completely multiplicative** if and only if

$$\forall a \cdot b = n, \quad f(n) = f(a) \cdot f(b)$$

and this multiplicativity also holds for every non-trivial decomposition of n. That is,

$$\forall n = w_1 \cdot w_2 \cdots w_k,\ 1 < w_i < n, \quad f(n) = f(w_1) \cdot f(w_2) \cdots f(w_k).$$

Since Koishi does not like multiplicativity, she proposes a **completely non-multiplicative** function $g(n)$, hoping that

$$\forall n = w_1 \cdot w_2 \cdots w_k,\ 1 < w_i < n, \quad g(n) \neq g(w_1) \cdot g(w_2) \cdots g(w_k).$$

To make the problem simpler, Koishi defines:

$$g(1) = 1,$$

$$g(n) = \operatorname{mex}\left(\left\{\prod g(w_i) \;\middle|\; n = \prod w_i,\ 1 < w_i < n\right\} \cup \{0\}\right), \quad n \geq 2,$$

where $\operatorname{mex}(S)$ denotes the smallest non-negative integer that does **not** appear in the set S.

Since Koishi is an unconscious girl, she asks you to compute the values of g. More precisely, you are given an integer N and a parameter c $(1 \leq N \leq 10^7,\ 1 \leq c < 998244353)$, and you need to output

$$\text{ans} = \left(\sum_{i=1}^{N} g(i) \cdot c^i\right) \bmod 998244353.$$

Input

The first line contains two integers N and c $(1 \leq N \leq 10^7,\ 1 \leq c < 998244353)$.

Output

Output a single integer — the answer.

Example

standard input	standard output
8 10	312121110
114 514	265738076

Note

Problem F. Four

Input file:	`standard input`
Output file:	`standard output`
Time limit:	4 seconds
Memory limit:	512 megabytes

Given four positive integers A, B, C, D and four non-negative integers p_1, p_2, p_3, p_4, you need to compute

$$\sum_{i=1}^{A}\sum_{j=1}^{B}\sum_{k=1}^{C}\sum_{l=1}^{D} i^{p_1} \cdot j^{p_2} \cdot k^{p_3} \cdot l^{p_4} \cdot \operatorname{lcm}\big(\gcd(i,j),\ \gcd(j,k),\ \gcd(k,l)\big)$$

modulo 998244353.

Here $\gcd(x, y)$ denotes the greatest common divisor of x and y, and $\operatorname{lcm}(x, y, z)$ denotes the least common multiple of x, y, z.

Input

The input contains a single line with eight integers, in order: $A, B, C, D, p_1, p_2, p_3, p_4$ ($1 \le A, B, C, D \le 2 \times 10^6$, $0 \le p_1, p_2, p_3, p_4 \le 10^9$).

Output

Output a single integer — the required sum modulo 998244353.

Example

standard input	standard output
`1 1 1 2 0 1 2 3`	`9`
`3 4 5 6 2 3 5 7`	`427375642`

Note

Problem G. Bipartite Graph

Input file:	`standard input`
Output file:	`standard output`
Time limit:	6 seconds
Memory limit:	1024 megabytes

Given a bipartite graph G with left vertex set L of size n and right vertex set R of size m. The edge set is E. The left vertices are numbered from 1 to n, and the right vertices are numbered from $n+1$ to $n+m$. Let $\mathcal{S}$ denote the set of all minimum vertex covers of G (each minimum vertex cover is a subset of the vertex set).

You are given q independent queries. Each query consists of several "requirements each of the form:

- u 0: vertex u must be selected (i.e., belongs to the considered minimum vertex cover);
- u 1: vertex u must not be selected (i.e., does not belong to the considered minimum vertex cover).

For each query, you need to determine whether there exists a minimum vertex cover $P \in \mathcal{S}$ that satisfies all the requirements of this query.

If it exists, you also need to compute:

1. Among all minimum vertex covers P that satisfy all requirements of this query, the number of vertices that are **forced to be selected** (i.e., the size of the intersection of all such minimum vertex covers);

2. Among all minimum vertex covers P that satisfy all requirements of this query, the number of vertices that are **forced not to be selected** (i.e., the size of the complement of the union of all such P, or equivalently, the number of vertices that do not appear in any of these P).

Input

The first line contains three integers n, m, e, q ($1 \le n, m \le 10^5$, $0 \le e \le \min(3\times10^5, nm)$, $1 \le q \le 2\times10^5$): the number of left vertices, the number of right vertices, the number of edges, and the number of queries.

The next e lines each contain two integers u, v, indicating an edge between left vertex u and right vertex v.

Then follow q queries. Each query is given in the following format:

The first line contains an integer t, the number of requirements in this query.

The next t lines each contain two integers x, y, where x is the vertex number, $y \in \{0, 1\}$. Here $y = 0$ means the vertex must be selected, and $y = 1$ means the vertex must not be selected.

The total number of requirements over all queries does not exceed 3×10^5.

Output

For each query, if no minimum vertex cover satisfies all the requirements, output a single line containing -1. Otherwise, output a line containing two integers: the number of vertices **forced to be selected** and the number of vertices **forced not to be selected**.

Example

standard input	standard output
3 5 15 5 1 4 1 5 1 6 1 7 1 8 2 4 2 5 2 6 2 7 2 8 3 4 3 5 3 6 3 7 3 8 2 1 0 4 1 2 1 0 1 0 2 3 1 8 0 2 6 1 1 1 2 8 0 8 0	3 5 3 5 -1 -1 -1

Note

Problem H. Count it

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 seconds
Memory limit:	256 megabytes

Given an integer sequence $A_1, A_2, \ldots, A_N$ of length N, where each element satisfies $0 \le A_i < M$.

For each i, define:

— Prefix set $S_i = \{A_1, A_2, \ldots, A_i\}$;

— Suffix set $T_i = \{A_{i+1}, A_{i+2}, \ldots, A_N\}$.

Let

$$P_i = \text{mex}(S_i), \qquad Q_i = \text{mex}(T_i), \qquad C_i = \min(P_i, Q_i),$$

where $\text{mex}(X)$ denotes the **minimum excluded** non-negative integer, i.e., the smallest non-negative integer x such that $x \notin X$.

Now you are given an array $B_1, B_2, \ldots, B_{N-1}$ of length $N-1$. You need to count how many sequences A of length N (with $0 \le A_i < M$) satisfy that for every $1 \le i < N$, $C_i = B_i$.

Output the answer modulo 998244353.

Input

The first line contains two integers N, M ($2 \le N \le 5000$, $1 \le M \le 10^{18}$).

The second line contains $N-1$ integers $B_1, B_2, \ldots, B_{N-1}$ ($0 \le B_i \le M$).

Output

Output a single integer — the number of valid sequences A modulo 998244353.

Example

standard input	standard output
4 2 1 2 1	1
5 2 1 1 2 1	1
10 10 0 0 0 0 6 0 0 0 0	0
10 100 0 1 2 2 2 2 2 2 0	279689447

Note

Problem I. Sequence Operation 2

Input file: `standard input`
Output file: `standard output`
Time limit: 1 seconds
Memory limit: 256 megabytes

Given an integer n, let $N = 2^n - 1$. You have a binary string s of length N, indexed from 1 to N.

Now perform N operations, numbered $x = 1, 2, \ldots, N$. For the x-th operation, you may choose one of the following two actions:

— **Skip**: do nothing;

— **Perform a flip**: choose a pair of integers (y, z) such that $1 \le y, z \le N$ and $x = y \oplus z$, where $\oplus$ denotes the bitwise XOR operation. Then flip the bits at positions x, y, and z simultaneously (i.e., change 0 to 1 and 1 to 0).

You need to provide a sequence of operations such that after all operations, the final binary string contains at most one position with value 1. In the output, if you perform operation x, you only need to specify y; the value of z is uniquely determined by $z = x \oplus y$. It is guaranteed that a solution always exists.

Input

The first line contains a single integer T ($1 \le T \le 1000$), the number of test cases.

Each test case consists of two lines:

The first line contains an integer n ($1 \le n \le 20$);

The second line contains a binary string s of length $N = 2^n - 1$, representing the initial binary string.

It is guaranteed that $\sum N \le 2 \times 10^6$ over all test cases.

Output

For each test case, output one line containing N integers $b_1, b_2, \ldots, b_N$, where b_x describes the x-th operation:

— If you skip the operation, set $b_x = 0$;

— Otherwise, let $y = b_x$ and $z = x \oplus y$, and perform the flip on positions x, y, z. In this case, it must hold that $1 \le y, z \le N$.

Example

standard input	standard output
8 2 000 2 100 2 010 2 110 2 001 2 101 2 011 2 111	0 0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 1 0 3 0 0 3 0

Note

Problem J. Sequence（Median Version）

Input file:	`standard input`
Output file:	`standard output`
Time limit:	2 seconds
Memory limit:	512 megabytes

Given a circular integer sequence $a_0, a_1, \ldots, a_{n-1}$ of length n. Define one **transformation** as follows:

1. Construct a new sequence $b_0, b_1, \ldots, b_{n-1}$ from the current sequence a, where

$$b_i = \text{median}\{a_i,\ a_{(i+1) \bmod n},\ a_{(i+n-1) \bmod n}\}.$$

2. Replace a by b.

Here $\text{median}(x, y, z)$ denotes the median of the three numbers, i.e., the value left after removing the maximum and the minimum.

Given the initial sequence a and the number of transformations k, please find the sequence after applying k transformations.

Input

The first line contains two integers n and k ($1 \le n \le 10^6$, $1 \le k \le 10^9$).

The second line contains n integers $a_0, a_1, \ldots, a_{n-1}$ ($0 \le a_i \le 10^9$), representing the initial sequence.

Output

Output a single line containing n integers — the final sequence, separated by spaces.

Example

standard input	standard output
3 1 0 1 2	1 1 1
4 1 0 1 2 3	1 1 2 2
4 2 0 2 0 2	0 2 0 2
10 4 0 1 2 3 4 2 1 0 5 6	1 1 2 3 3 2 1 1 5 5
12 6 12 9 7 8 1 2 4 5 6 7 14 20	12 9 8 7 2 2 4 5 6 7 14 14

Note

Problem K. Sequence（Mex Version）

Input file: standard input
Output file: standard output
Time limit: 2 seconds
Memory limit: 256 megabytes

Given a circular integer sequence $a_0, a_1, \ldots, a_{n-1}$ of length n. Define one **transformation** as follows:

1. Construct a new sequence $b_0, b_1, \ldots, b_{n-1}$ from the current sequence a, where

$$b_i = \operatorname{mex}\{a_i,\ a_{(i+1) \bmod n},\ a_{(i+n-1) \bmod n}\}.$$

Here mex denotes the minimum non-negative integer that does not appear in a set of non-negative integers (e.g., $\operatorname{mex}\{0, 1, 3\} = 2$).

2. Replace a by b.

Given the initial sequence a and the number of transformations k, please find the sequence after applying k transformations.

Input

The first line contains two integers n and k ($1 \le n, k \le 10^6$).

The second line contains n integers $a_0, a_1, \ldots, a_{n-1}$ ($0 \le a_i \le 10^9$), representing the initial sequence.

Output

Output a single line containing n integers — the final sequence, separated by spaces.

Example

standard input	standard output
4 1 0 1 2 3	2 3 0 1
4 2 0 2 0 2	0 0 0 0

Note

Problem L. Matrix

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 seconds
Memory limit:	1024 megabytes

Given an integer matrix a of size $n \times m$.

You may perform the following operation any number of times: choose a path from $(1, 1)$ to (n, m), moving only right or down at each step, and then increase by 1 every cell on that path.

Your goal is to make all elements of the matrix equal.

Find the minimum number of operations required; if it is impossible, output -1.

Input

The first line contains a single integer T ($1 \le T \le 5000$), the number of test cases.

Each test case is given in the following format:

The first line contains two integers n, m, the number of rows and columns.

The next n lines each contain m integers, representing the matrix a, where the element at row i, column j is $a_{i,j}$ ($-10^9 \le a_{i,j} \le 10^9$).

It is guaranteed that $1 \le \sum n, \sum m \le 5000$ over all test cases.

Output

For each test case, output a single integer on a separate line: the minimum number of operations needed. If it is impossible to make all numbers equal, output -1.

Example

standard input	standard output
2 2 3 1 1 1 2 2 1 3 2 1 2 2 2 2 1	1 2

Note

Problem M. Je t'aime encore

Input file:	`standard input`
Output file:	`standard output`
Time limit:	5 seconds
Memory limit:	512 megabytes

Je t'aime encore
Mais le monde pleure sans fin

— MONEY, 《*Je t'aime encore*》

A full ternary plane tree is a rooted tree with leaves, where every internal node has exactly 3 children.

Define the weight of each node as follows:

1. The weight of a leaf is 0;

2. For an internal node, if the weights of its three children are a, b, c, let $m = \max\{a, b, c\}$. If the value m appears at least twice among a, b, c, then the weight of this node is $m + 1$; otherwise, its weight is m.

Now you are given integers H, N, K.

Consider all full ternary plane trees satisfying the following conditions:

1. The weight of the root is exactly H;

2. The number of internal nodes is exactly $2^H - 1 + N$.

For each such tree, colour all internal nodes whose weight is exactly H blue.

For each $1 \le k \le K$, you need to compute the number of valid trees that have exactly k blue nodes, modulo 2.

Input

The first line contains a single integer T ($1 \le T \le 5$), the number of test cases.

Each of the next T lines contains three integers H, N, K ($1 \le H \le 10^{18}$, $1 \le K \le N + 1$), as described above.

It is guaranteed that $0 \le \sum N \le 5 \times 10^5$ over all test cases.

Output

For each test case, output a single line consisting of a string of length K of '0' and '1'.

The k-th character ($1 \le k \le K$) represents the number of trees with exactly k blue nodes modulo 2 (i.e., output '1' if the count is odd, otherwise '0').

Example

standard input	standard output
1 2 10 11	01110001011
1 3 3 4	0011
1 20 500000 17	00000000000000000

Note

Problem N. Welcome to HIT

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 seconds
Memory limit:	256 megabytes

Many universities are commonly referred to by uppercase English abbreviations. For example, Tsinghua University is abbreviated as THU, and Peking University is abbreviated as PKU.

The English name of 哈尔滨工业大学 is Harbin Institute of Technology. Please output its uppercase English abbreviation.

Input

There is no input for this problem.

Output

Output the uppercase English abbreviation of Harbin Institute of Technology.

Example

standard input	standard output
—	THU

Note

The sample output only demonstrates the required output format and is not necessarily the correct answer to this problem.